

Multimodal Inputs

- **Joint Kinematics (18 ch):**

Hip, Knee, Ankle 3D (deg)

Angular Vel. $\dot{\mathbf{q}}$ (rad/s)

- **Kinetic Loads (6 ch):**

3D Bilateral GRF (N/kg)

- **Musculotendon State (80 ch):**

Fiber Length l_m (40 ch)

Fiber Velocity v_m (40 ch)

- **Unified Sensor Dimension:**

Input Vector $\mathbf{x}_t \in \mathbb{R}^{102}$

102

Projection

- **Linear Dense:**

$\mathbb{R}^{102} \rightarrow \mathbb{R}^{64}$

- **Regularization:**

Dropout ($p = 0.1$)

- **Normalization:**

LayerNorm $\mathbb{R}^{64}$

- **Embed Token:**

$\mathbf{u}_t \in \mathbb{R}^{64}$

64

Stacked S4D Backbone ($\times 4$ Blocks, $d_{\text{model}} = 64$)

+ Skip Residual

Diagonal S4D

- **Continuous SSM:**

$\mathbf{h} = \mathbf{A}\mathbf{h} + \mathbf{B}\mathbf{x}$

- **Discretization:**

Zero-Order Hold

- **HiPPO State:**

$d_{\text{state}} = 64$

- **Diag Kernel K:**

$\mathcal{O}(N)$ FFT Conv

FiLM

$\odot, +$

$\mathbf{z}_i, \tilde{\mathbf{z}}_i$

Feed-Forward

- **Non-linearity:**

GELU Activation

- **Regularization:**

Dropout ($p = 0.1$)

- **Projection:**

Dense $\mathbb{R}^{64} \rightarrow \mathbb{R}^{64}$

- **Residual Add:**

$\mathbf{x} + \tilde{\mathbf{z}}_i$ & Norm

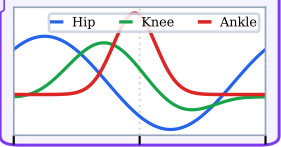
Predicted Joint Kinetics

- **LayerNorm + Projection Head:**

$\mathbb{R}^{64} \rightarrow \mathbb{R}^3$ Sagittal Moments

- **Predicted Torques $\hat{\boldsymbol{\tau}}(t)$:**

Sagittal Kinetics (0-100% Gait)



0% 50% 100%

Multi-Task Loss $\mathcal{L}$

Physics Regularization

- **Supervised Tracking Loss:**

$\mathcal{L}_{\text{task}} = \text{MSE}(\hat{\boldsymbol{\tau}}, \boldsymbol{\tau}^*)$

- **Torque Jerk Penalty:**

$\mathcal{L}_{\text{jerk}} = \frac{1}{3T} \sum \left(\frac{d^2 \hat{\tau}_j}{dt^2} \right)^2$

- **Power Consistency Loss:**

$\mathcal{L}_{\text{power}} = \frac{1}{3T} \sum_{t,j} (\hat{\tau}_j \dot{q}_j - P_j^*)^2$

Speed Context

- **Treadmill Velocity s :**

Continuous: 0.6-1.6 m/s

- **Normalized Scalar:**

Replicated across $T = 100$

- **Conditioning Driver:**

Gait Dynamics Modulation

Speed s [B, 1]

FiLM Conditioning Generator (Block l)

- **2-Layer Speed Adaptation MLP (SiLU):**

$s \in \mathbb{R} \rightarrow \text{Linear}(1 \rightarrow 32) \rightarrow \text{SiLU}$

$\rightarrow \text{Linear}(32 \rightarrow 128) \rightarrow (\boldsymbol{\gamma}_l, \boldsymbol{\beta}_l) \in \mathbb{R}^{2d_{\text{model}}}$

- **Affine Transformation Applied to State $\mathbf{z}_i$:**

$\tilde{\mathbf{z}}_i = (1 + \boldsymbol{\gamma}_l) \odot \mathbf{z}_i + \boldsymbol{\beta}_l$

- **Zero Initialized for Identity Bias:**

$\mathbf{W}_{l,2} = \mathbf{0}, \mathbf{b}_{l,2} = \mathbf{0} \Rightarrow \tilde{\mathbf{z}}_i = \mathbf{z}_i$ at init

$(\boldsymbol{\gamma}_l, \boldsymbol{\beta}_l) \in \mathbb{R}^{64 \times 2}$